

ASWIN P T

Software Developer

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PROFESSIONAL SUMMARY:

Machine Intelligence graduate student at Digital University Kerala with strong foundations in AI/ML, deep learning, and computer vision. Experienced in implementing research papers, building end-to-end ML pipelines, and optimizing models for real-world performance. Skilled in Python, PyTorch, OpenCV, and GPU-accelerated development. Demonstrated ability to work with complex datasets, experiment systematically, and derive meaningful insights. Driven by discipline, consistency, and a strong research mindset.

EDUCATION:

M.Sc. in Computer Science (Machine Intelligence)

Kerala University of Digital Sciences, Innovation and Technology, Thiruvananthapuram (2024 – 2026)

B.Sc. in Computer Science

College of Applied Science IHRD, Kozhikode (2021– 2024)

TECHNICAL SKILLS:

- **Programming:** Python, C, C++
- **AI/ML:** Deep Learning, Computer Vision, Federated Learning, Data Analytics, Cognitive Computing
- **Frameworks:** PyTorch, TensorFlow, OpenCV
- **Tools:** Google Colab, Kaggle, VS Code, PyCharm, Jupyter Notebook
- **Databases:** PostgreSQL, SQL
- **Web:** HTML, CSS, JavaScript, Django

PROJECTS:

Adaptive Confidence Modeling for SHADER-Based 3D Human Body Shape Estimation (Ongoing)

Python, PyTorch, OpenCV, SMPL

Developed an improved SHADER pipeline for monocular 3D human shape estimation under loose clothing by introducing a dynamic, image-driven confidence model to replace the original global- σ formulation. Achieved 20–25% lower confidence-prediction error on 9,656 dressed-human images with statistically significant gains ($p < 0.000001$), producing more stable confidence maps and improving downstream SMPL estimation. The approach is directly applicable to virtual try-on systems, avatar generation, AR/VR human modeling, and digital fashion applications without requiring additional data or architectural changes.

Deep Learning-Based Prediction of RNA Secondary Structures

Python, CNN, RNN

Designed a model to predict RNA secondary structures from sequences. Achieved good training accuracy and F1-score on validation using dot-bracket notation.

PM2.5 Air Quality Forecasting

Python, TensorFlow, Pandas, Seaborn

Developed a TensorFlow-based LSTM model to forecast 7-day PM2.5 levels in Mumbai, New Delhi, Hyderabad, and Chennai. Applied data preprocessing techniques including imputation, normalization, and feature encoding. Conducted exploratory data analysis and visualized pollution patterns.

Manglish WhatsApp Chatbot using Llama-3 (Ollama) and Node.js

Built an offline Manglish (Malayalam–English) WhatsApp chatbot using a hybrid AI architecture combining rule-based intent analysis, similarity-based response retrieval, and local Llama-3 inference via Ollama. Designed intent patterns and conversational fallbacks to handle informal, low-resource native language input, enabling cloud-independent, real-time messaging and a scalable foundation for regional language chatbots.